The documentation and process conversion measures necessary to comply with this document shall be completed by 16 April 2006

INCH-POUND

MIL-PRF-19500/560G
16 January 2006
SUPERSEDING
MIL-PRF-19500/560F
27 May 2004

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING,
TYPE 2N5339 AND 2N5339U3, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon switching transistors. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500 and two levels of product assurance are provided for each unencapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-39), figure 2 for U3 devices (TO-276AA) and figures 3, 4, and 5 for JANHC and JANKC (die) dimensions.

1.3 Maximum ratings Unless specified, T_A = 25°C.

Types	P_T (1) T_A = +25°C	P_T (1) T_C = +25°C	$R_{\theta JA}$	$R_{\theta JC}$	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_J and T_{STG}
	W	W	°C/W	°C/W	V dc	V dc	V dc	A dc	A dc	°C
2N5339	1.0	17.5	175	10	100	100	6.0	5.0	1.0	-65 to +200
2N5339U3	1.0	75		2.3	100	100	6.0	5.0	1.0	-65 to +200

(1) For derating, see figures 6, 7, and 8.

1.4 Primary electrical characteristics T_A = +25°C. (Unless otherwise indicated, applies to all devices.)

Limits	h_{FE1} (1) V_{CE} = 2.0 V dc; I_C = 0.5 A dc	h_{FE2} (1) V_{CE} = 2.0 V dc; I_C = 2.0 A dc	h_{FE3} (1) V_{CE} = 2.0 V dc; I_C = 5.0 A dc
Min	60	60	40
Max		240	

(1) See note at end of 1.4.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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1.4 Primary electrical characteristics T_A = +25°C. - Continued.

Limits	$\|h_{FE}\|$	C_{obo}	Switching		$V_{CE(SAT)1}$	$V_{BE(SAT)1}$
	f = 10 MHz V_{CE} = 10 V dc I_C = 0.5 A dc	V_{CE} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	t_{on}	t_{off}	I_C = 2.0 A dc I_B = 0.2 A dc (1)	I_C = 2.0 A dc I_B = 0.2 A dc (1)
		pF	µs	µs	V dc	V dc
Min	3.0					
Max	15.0	250	0.2	2.2	0.7	1.2

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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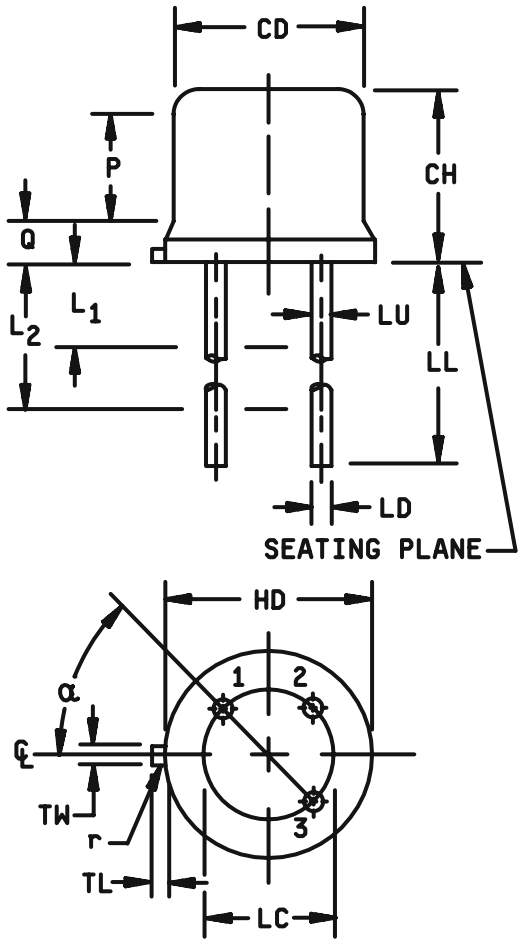


FIGURE 1. Physical dimensions (TO-39).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.355	7.75	9.02	5
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	3
LC	.200 TP		5.08 TP		6
LD	.016	.021	.41	.53	7
LL	.500	.750	12.70	19.05	7
LU	.016	.019	.41	.48	7
L1		.050		1.27	7
L2	.250		6.35		7
TL	.029	.045	.74	1.14	3
TW	.028	.034	.71	.86	10
P	.100		2.54		5
Q		.050		1.27	4
r		.010		.25	10, 11
α	45° TP		45° TP		6
Notes	1, 2, 8, 9		1, 2, 8, 9		

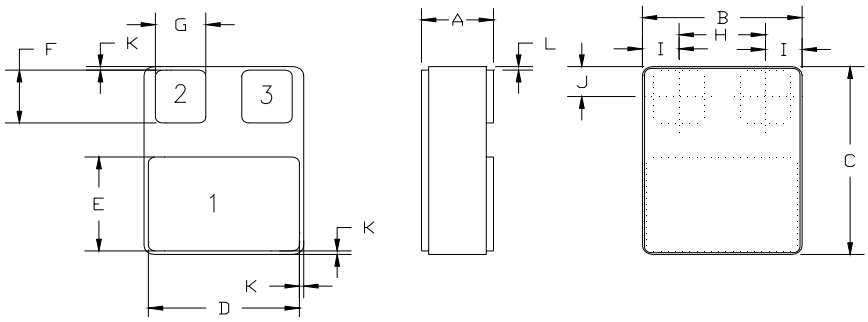
NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Symbol CD shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) relative to tab. Device may be measured by direct methods or by gauge.
7. Symbol LD applies between L1 and L2. Dimension LD applies between L2 and LL minimum.
8. Lead designation, depending on device type, shall be as follows:

Lead number	TO-39
1	Emitter
2	Base
3	Collector

9. Lead number three is electrically connected to case.
10. Beyond r maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
11. Symbol r applied to both inside corners of tab.
12. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (TO-39) - Continued.

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BOTTOM VIEW

Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.111	.122	2.82	3.10
B	.291	.301	7.39	7.65
C	.395	.405	10.03	10.29
D	.281	.291	7.14	7.39
E	.220	.230	5.59	5.84
F	.115	.125	2.92	3.18
G	.09	.100	2.29	2.54
H	.145	.155	3.68	3.94
I	.073 TYP.		1.85 TYP.	
J	.083 TYP.		2.11 TYP.	
K	.005 TYP.		0.13 TYP.	
L	.015 TYP.		0.38 TYP.	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

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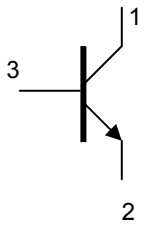
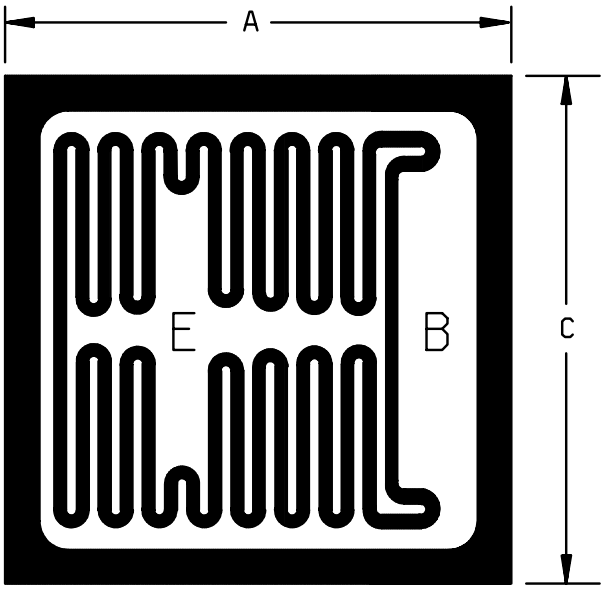


FIGURE 2. Physical dimensions and configuration (U3) (SMD 5) (TO-276AA).

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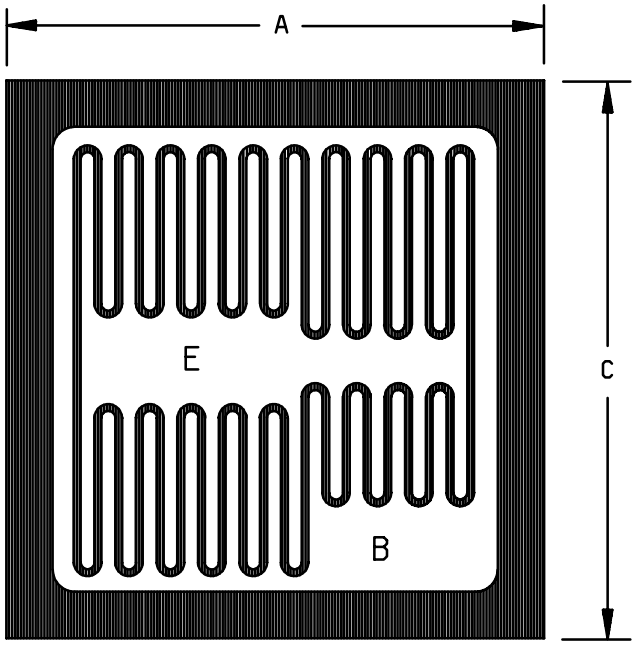


Letter	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.120	.124	3.05	3.15
C	.120	.124	3.05	3.15

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .008 inch (0.20 mm) to .012 inch (0.30 mm).
   Top metal: Aluminum 40,000 Å minimum, 50,000 Å nominal.
   Back metal: Gold 2,500 Å minimum, 3,000 Å nominal.
   Back side: Collector.
   Bonding pad: B = .015 inch (0.38 mm) x .072 inch (1.83 mm).
   E = .015 inch (0.38 mm) x .060 inch (1.52 mm).
4. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm).
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions JANHCA and JANKCA.

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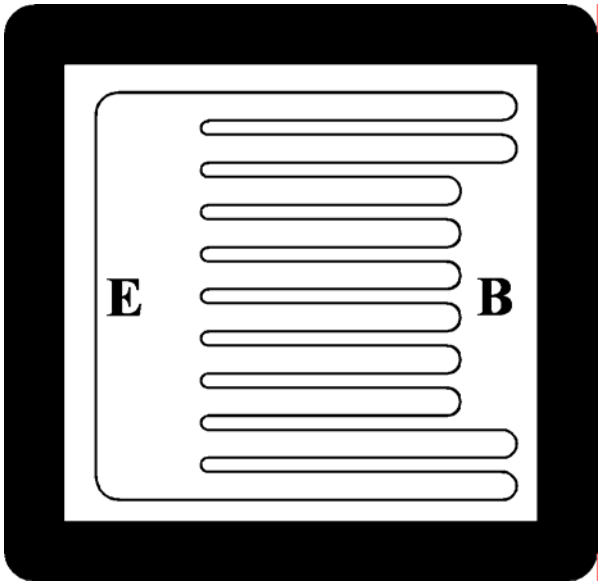


Letter	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.098	.102	2.49	2.59
C	.098	.102	2.49	2.59

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .006 inch (0.15 mm) to .010 inch (0.25 mm).
   Top metal: Aluminum 25,000 Å minimum, 37,500 Å nominal.
   Back metal: Gold 1,500 Å minimum, 6,500 Å nominal.
   Back side: Collector.
4. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm).
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Physical dimensions JANHCB and JANKCB.

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NOTES:

1. Chip size. .120 (3.05 mm) x .120 (3.05 mm) inch ±.002 inch (0.051 mm).
2. Chip thickness .010 (0.254 mm) ±.0015 (0.038 mm) inch nominal.
3. Top metal Aluminum 30,000Å minimum, 33,000Å nominal.
4. Back metal A. Al/Ti/Ni/Ag12kÅ/3kÅ/7kÅ/7kÅ minimum15kÅ/5kÅ/10kÅ/10kÅ nominal.
   B. Gold 2,500Å minimum, 3000Å nominal.
5. Backside Collector.
6. Bonding pad B = .052 inch (1.32 mm) x .012 inch (0.305 mm).
   E = .084 inch (2.13 mm) x .012 inch (0.305 mm).
7. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 5. Physical dimensions JANHCC and JANKCC.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO-39), figure 2 for U3 (TO-276AA), and figures 3, 4, and 5 (JANHC and JANKC) devices herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I, group A herein.

3.7 Marking. Devices shall be marked in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.2.1.1 Group E thermal impedance. Each supplier shall submit a thermal impedance ($Z_{\theta JX}$) histogram of the entire qualification lot. The histogram data shall be taken prior to the removal of devices that are atypical for thermal impedance. Thermal impedance curves (from $Z_{\theta JX}$ test pulse time to $R_{\theta JX}$ minimum steady-state time) of the best device in the qual lot and the worst device in the qual lot (that meets the supplier proposed screening limit), or from the thermal grouping, shall be submitted. The optimal test conditions and proposed initial thermal impedance screening limit shall be provided in the qualification report. Data indicating how the optimal test conditions were derived for $Z_{\theta JX}$ shall also be submitted. The proposed specification maximum thermal impedance curve shall be submitted. The qualifying activity may approve a different $Z_{\theta JX}$ limit not to exceed the specification's thermal curve for conformance inspection end-point measurements as applicable**.** Equivalent data, procedures, or statistical process control plans may be used for part, or all, of the above requirements. The approved thermal impedance conditions and limit for $Z_{\theta JX}$ shall be used by the supplier in screening and table I, subgroup 2. The approved thermal resistance conditions for $R_{\theta JX}$ shall be used by the supplier for conformance inspection. For product families with similar thermal characteristics based on the same physical and thermal die, package, and construction combination (thermal grouping), the supplier may use the same thermal impedance curves.

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4.2.2 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

* 4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see MIL-PRF-19500)	Measurements	
	JANS level	JANTX and JANTXV levels
*(1) 3c	Thermal impedance, method 3131 of MIL-STD-750.	Thermal impedance, method 3131 of MIL-STD-750.
7	Optional	Optional
9	I_{CBO1} and h_{FE2}	Not applicable
11	I_{CBO1}; h_{FE2}, ΔI_{CBO1} = ± 100 percent of initial value or 1.0 μA dc, whichever is greater; Δh_{FE2} = ± 15 percent	I_{CBO1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = ± 100 percent of initial value or 1.0 μA dc, whichever is greater; Δh_{FE2} = ± 15 percent.	Subgroup 2 of table I herein; ΔI_{CBO1} = ± 100 percent of initial value or 1.0 μA dc, whichever is greater; Δh_{FE2} = ± 15 percent.
14	Required	Required

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. Power shall be applied to the device to achieve T_J = minimum +175°C and minimum power dissipation of P_D = 75 percent P_T maximum as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions.) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Screening for JANHC and JANKC. Screening for JANHC and JANKC die shall be in accordance with MIL-PRF-19500, "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). Measurement delay time (t_{MD}) = 70 μs max. See table II, subgroup 4 herein.

* 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of subgroup 1 and 2, of table I herein, inspection only (table VIb, group B, subgroup 1 is not required to be performed since solderability and resistance to solvents testing is performed in subgroup 1 of table I herein.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa of MIL-PRF-19500 (JANS) and 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. See 4.4.2.2 herein for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein.

* 4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	V_{CB} = 10 V dc, 2,000 cycles, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
B5	1027	(NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) V_{CB} = 10 V dc; $P_D \geq$ 100 percent of rated P_T (see 1.3). Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table VIa, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve T_J = +225°C minimum.
B5	2037	Test condition A.

* 4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and /or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	n = 45, c = 0. Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High- temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

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4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

* 4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, N/A for U3 devices.
C5	3131	See 4.3.3.
C6	1037	For solder die attach: $V_{CB} \geq 10$ V dc, T_A = room ambient as defined in the general requirements of MIL-STD-750. 6,000 cycles. For JANS only.
* C6	1027	For JANS only. For eutectic die attach: $V_{CB} \geq 10$ V dc, adjust P_T to achieve T_J = + 175°C min.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500, and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
* Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 4/	1071	n = 22 devices, c = 0				
Fine leak Gross leak						
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 device, c = 0				
Subgroup 2						
Thermal impedance	3131	See 4.3.3	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 50 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	100		V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 100 V dc	I_{CEO}		100	µA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc; V_{CE} = 90 V dc	I_{CEX1}		1	µA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 100 V dc	I_{CBO}		1	µA dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Emitter to base, cutoff current	3061	Bias condition D; V_{EB} = 6.0 V dc	I_{EBO}		100	µA dc
Forward - current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 0.5 A dc, pulsed (see 4.5.1)	h_{FE1}	60		
Forward - current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 2.0 A dc; pulsed (see 4.5.1)	h_{FE2}	60	240	
Forward - current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 5.0 A dc; pulsed (see 4.5.1)	h_{FE3}	40		
Collector to emitter voltage (saturated)	3071	I_C = 2.0 A dc; I_B = 0.2 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)1}$		0.7	V dc
Collector to emitter voltage (saturated)	3071	I_C = 5.0 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)2}$		1.2	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 2.0 A dc; I_B = 0.2 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)1}$		1.2	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 5.0 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)2}$		1.8	V dc
Subgroup 3						
High - temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 90 V dc; V_{BE} = 1.5 V dc;	I_{CEX2}		1.0	mA dc
Low-temperature operation		T_A = -55°C				
Forward - current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 2.0 A dc; Pulsed (see 4.5.1)	h_{FE4}	12		

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small - signal short - circuit forward - current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 0.5 A dc; f = 10 MHz	$\|h_{fe}\|$	3	15	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		250	pF
Input capacitance (output open - circuited)	3240	V_{BE} = 2.0 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz (see 4.5.2)	C_{ibo}		1,000	pF
Pulse response						
Pulse delay time	3251	See figure 12	t_d		100	ns
Pulse rise time	3251	See figure 12	t_r		100	ns
Pulse storage time	3251	See figure 13	t_s		2	μs
Pulse fall time	3251	See figure 13	t_f		200	ns
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t ≥ 0.5 s; 1 cycle				
Test 1		V_{CE} = 2.0 V dc; I_C = 5.0 A dc				
Test 2		V_{CE} = 5.0 V dc; I_C = 2.0 A dc				
Test 3		V_{CE} = 90 V dc; I_C = 55 mA dc				
End-point electrical measurements		See table I, subgroup 2				
Subgroups 6 and 7						
Not applicable						

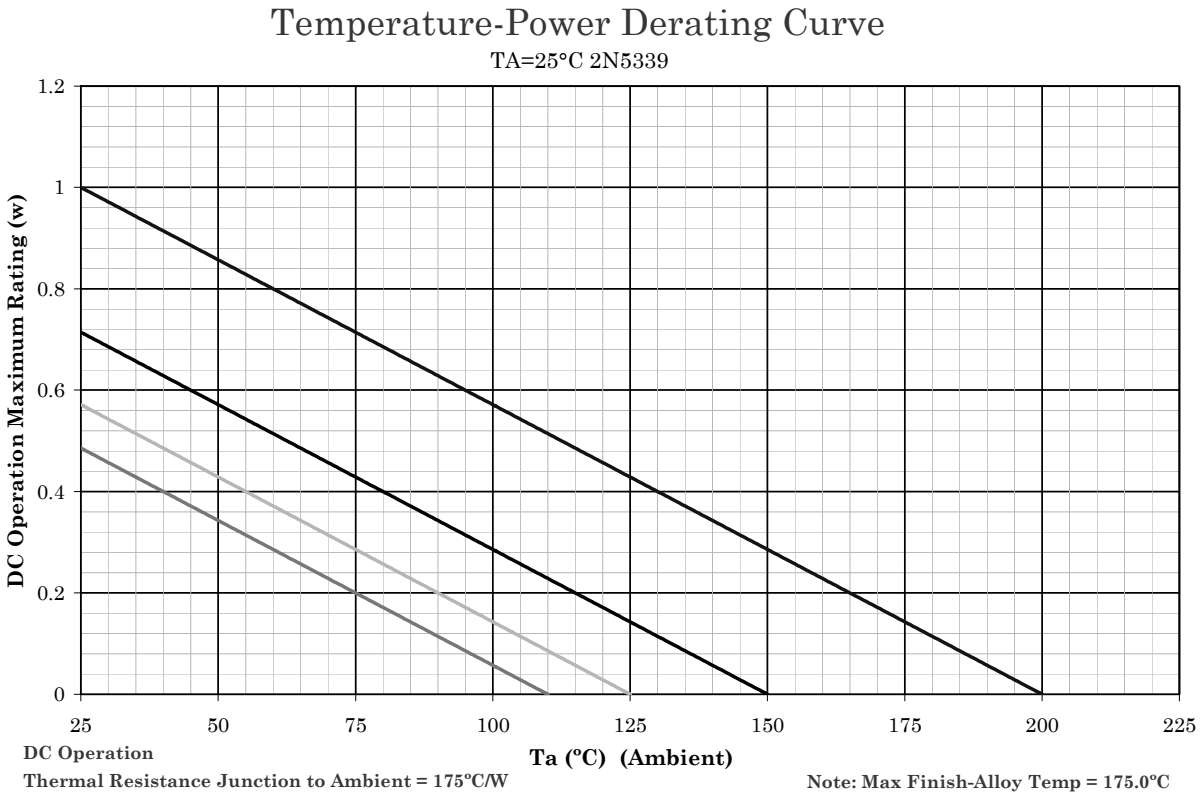
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.

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* TABLE II. Group E inspection (all quality levels) - for qualification and requalification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal			
Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroups 4			N/A
Thermal impedance curves		See 4.2.1.1.	
Subgroups 5 and 6			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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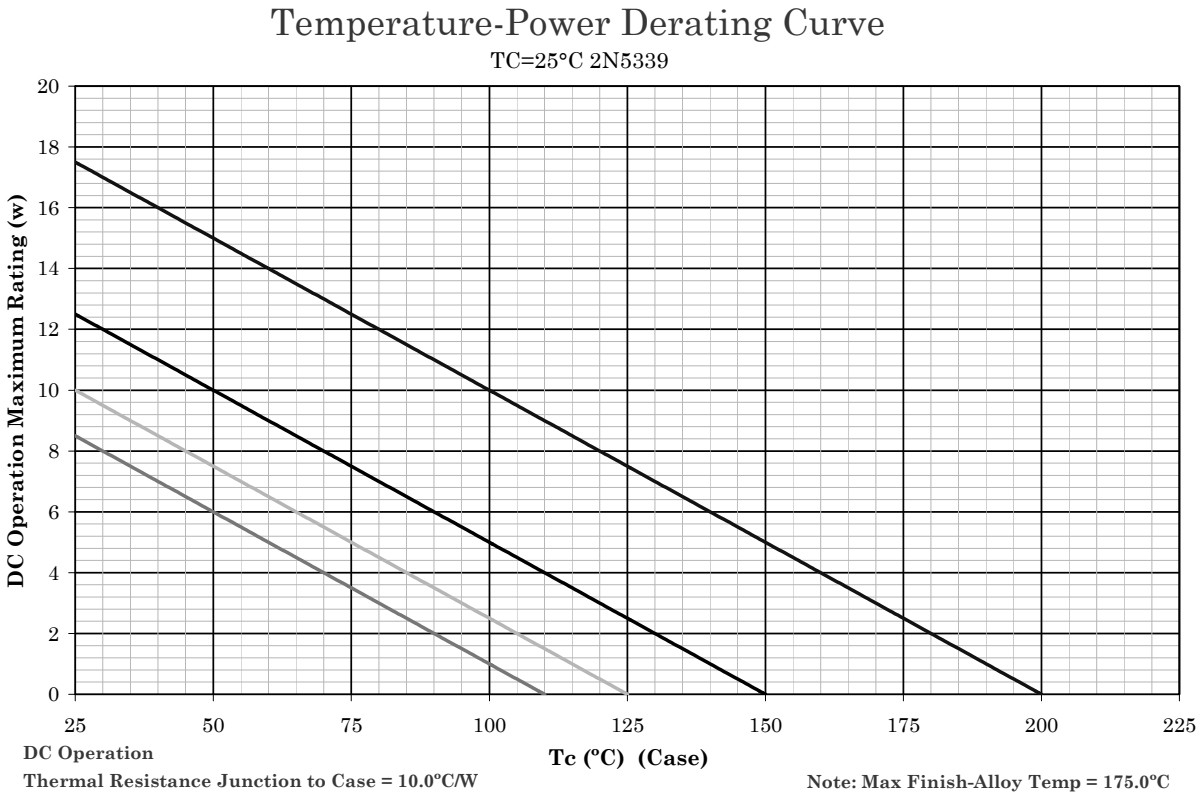


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 6. Derating for 2N5339 ($R_{\theta JA}$) (TO-39).

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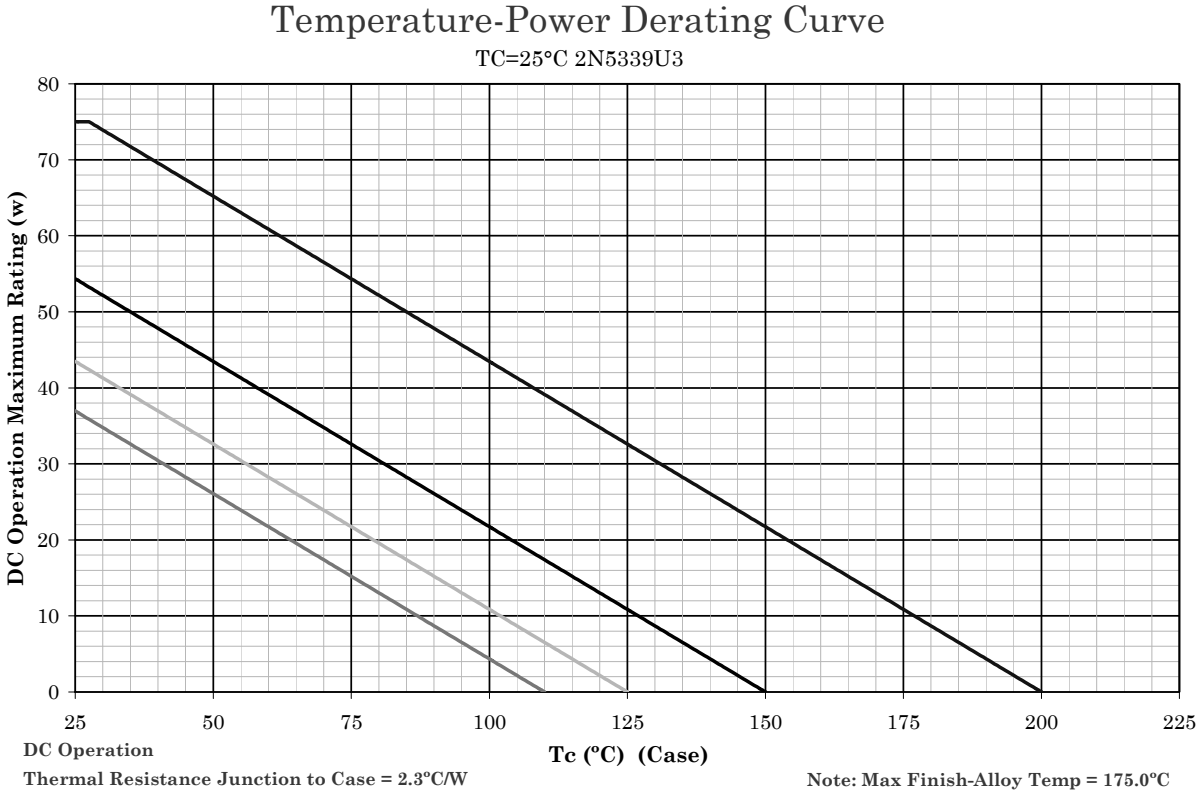


NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 7. Derating for 2N5339 ($R_{\theta JC}$) (TO-39).

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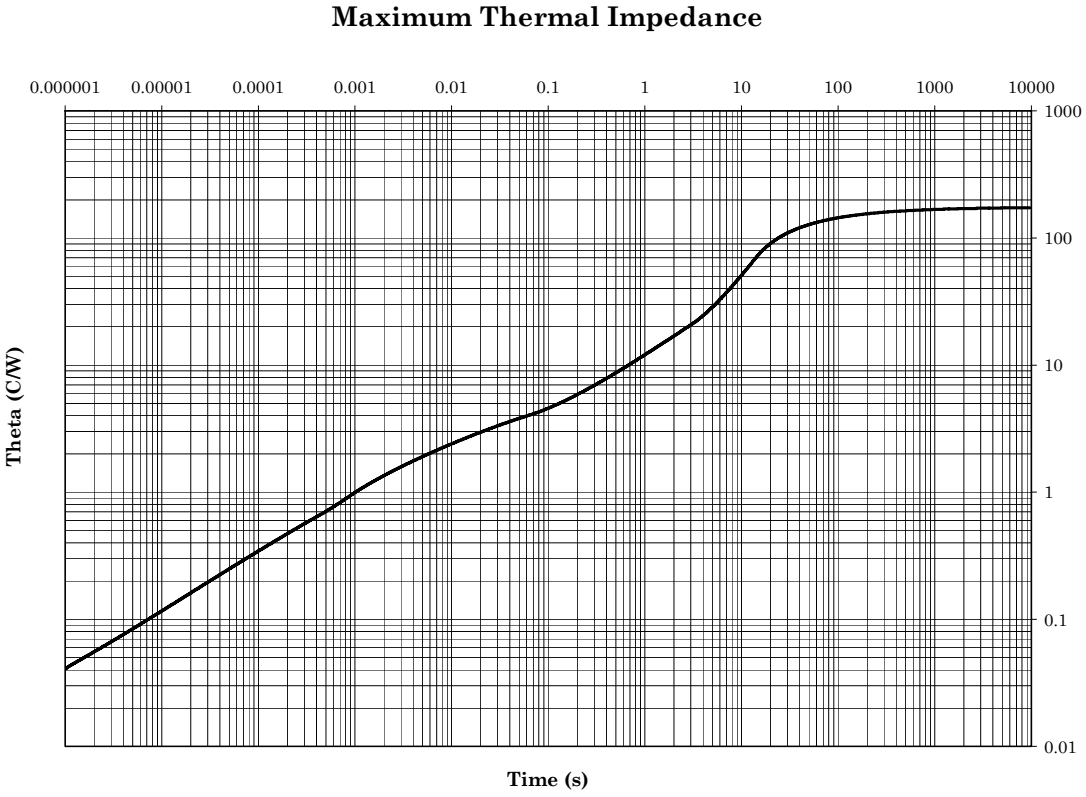


NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 8. Derating for 2N5339U3 ($R_{\theta JC}$).

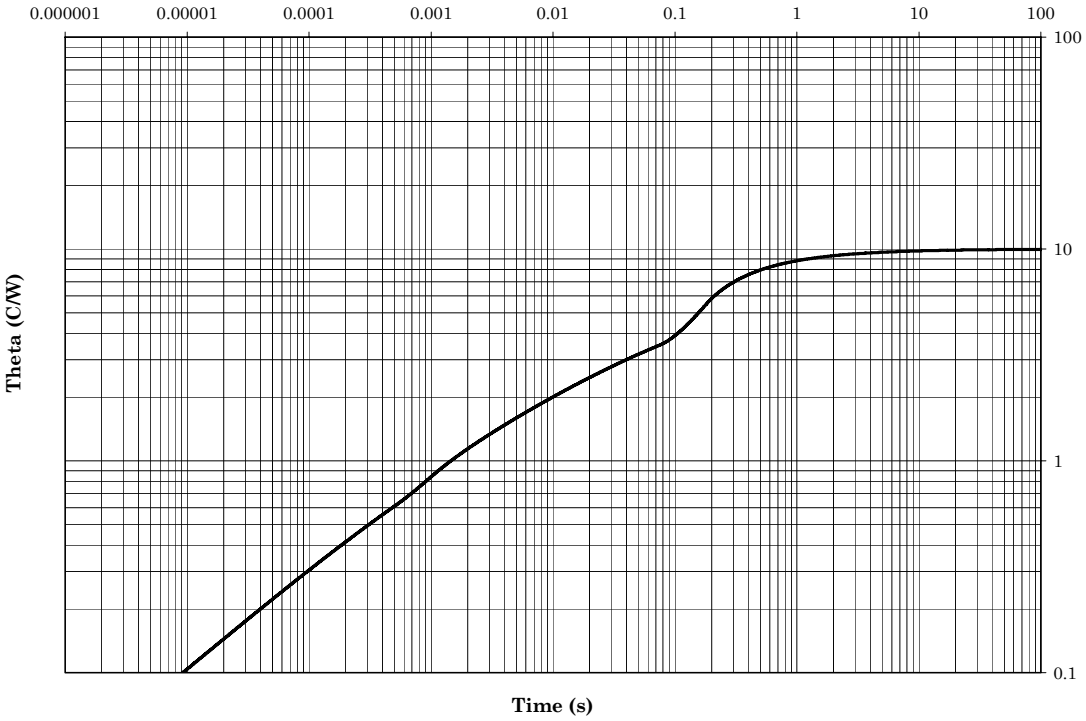
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T_A = +25°C, Pdiss = 1 W, thermal resistance = 175°C/W (ambient thermal resistance varies with power).

* FIGURE 9. Thermal impedance graph ($R_{\theta JA}$) for 2N5339 (TO-39).

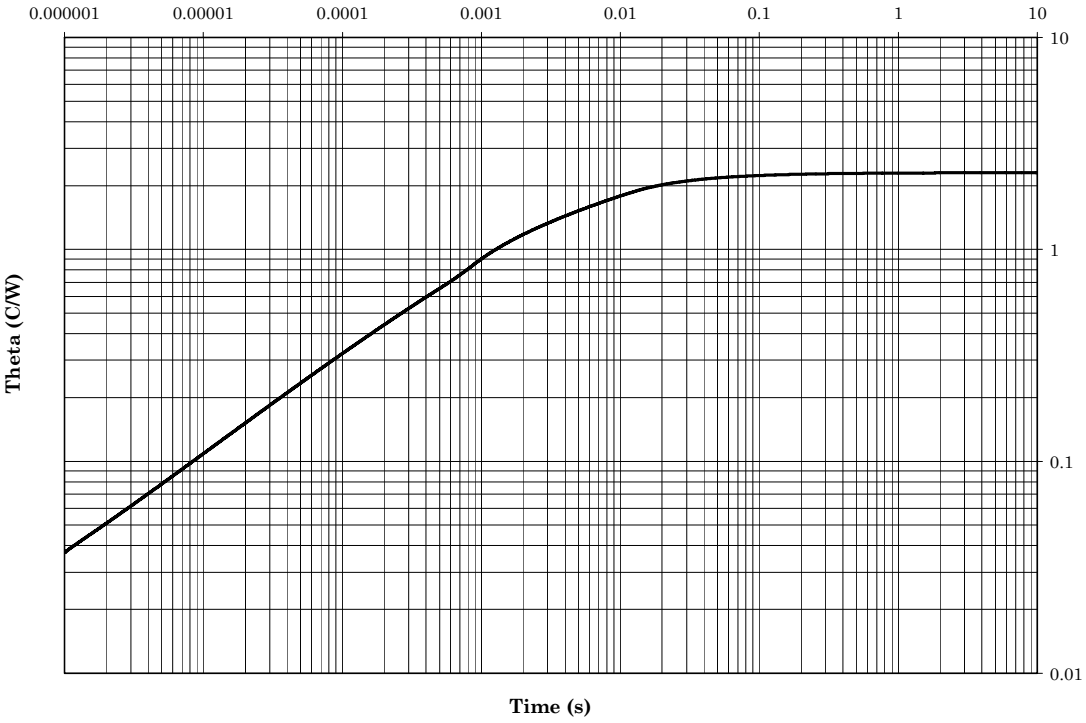
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T_C = +25°C, thermal resistance = 10°C/W.

* FIGURE 10. Thermal impedance graph ($R_{\theta JC}$) for 2N5339 (TO-39).

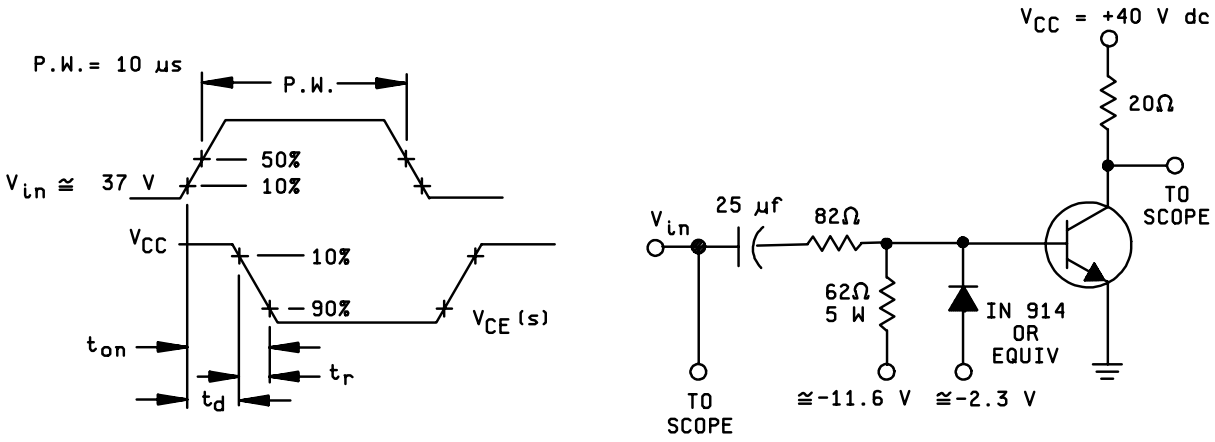
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T_C = +25°C, thermal resistance = 2.3°C/W.

* FIGURE 11. Thermal impedance graph ($R_{\theta JC}$) for 2N5339U3 (U3).

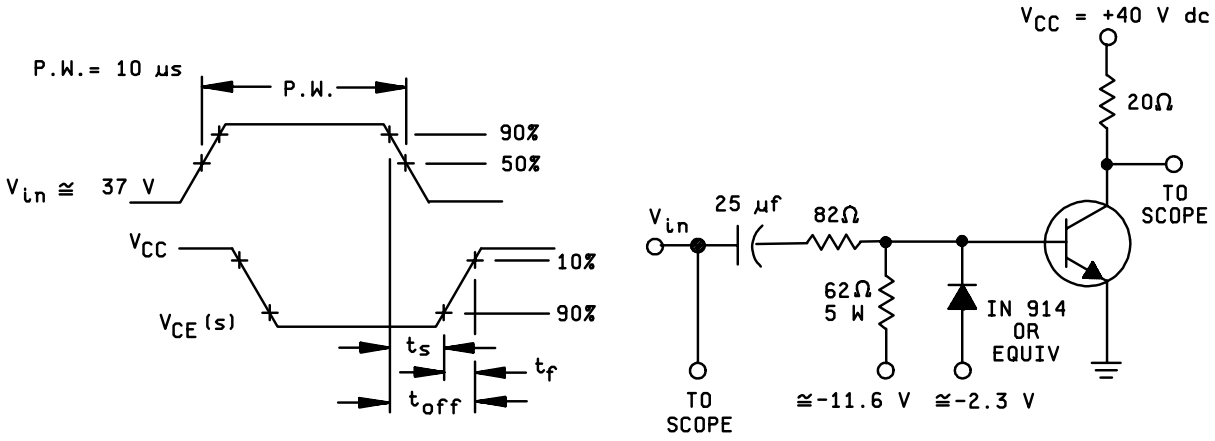
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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50Ω.
2. Sampling oscilloscope: Z_{in} ≥ 1 MΩ, C_{in} ≤ 20 pF, rise time ≤ 20 ns.
3. t_{on} conditions: I_C = 2 A, I_{B1} = 200 mA.

FIGURE 12. Saturated turn-on switching waveform and time test circuit.



NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50Ω.
2. Sampling oscilloscope: Z_{in} ≥ 1 MΩ, C_{in} ≤ 20 pF, rise time ≤ 20 ns.
3. t_{on} conditions: I_C = 2 A, I_{B1} = I_{B2} = 200 mA.

FIGURE 13. Saturated turn-off switching time waveform and test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 3 and 4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Application guidance. The following PNP type transistor is complimentary to the NPN device listed herein.

NPN	PNP
2N5339	2N6193

6.5 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example, JANHCA2N6193) will be identified on the QML.

JANC ordering information			
PIN	Manufacturers		
	33178	34156	43611
2N5339	JANHCA2N5339 JANKCA2N5339	JANHCB2N5339 JANKCB2N5339	JANHCC2N5339 JANKCC2N5339

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6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue

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Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2005-018)

Review activities:
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